

DRAFT — Petroleum Sector Framework: Design Basis and Military Action Extension

Design Basis and Military Action Extension

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Eclectic Technologies

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Preface

The petroleum sector occupies a foundational position in modern industrial civilization. Crude oil, refined into gasoline, diesel, jet fuel, heating oil, and the feedstocks for petrochemical production, powers transportation, military logistics, agricultural supply chains, and the manufacturing base on which every other sector depends. Yet the sector that underlies all others operates without a formal Design Basis — without a documented, methodologically rigorous statement of the bounding events its facilities must be designed to survive.

This document fills that gap and immediately extends it. The DBA-OIL sector framework establishes a Design Basis for the petroleum sector — a structured methodology for identifying bounding events, defining consequence envelopes, and specifying the system response standards that constitute adequate protection. It then extends that basis to the military action threat tier, asking what happens when the bounding event is not an equipment failure, a natural hazard, or an operator error, but rather a deliberate, sophisticated, state-directed attack against petroleum infrastructure.

The evidentiary record is unambiguous. Russian airstrikes on Ukrainian petroleum refining and storage infrastructure in 2022 through 2024 systematically destroyed refinery capacity, disrupted fuel supply for civilian and military consumers, and forced the Ukrainian economy to absorb the cascading consequences of domestic refinery loss. The Islamic Revolutionary Guard Corps has targeted Saudi Aramco facilities with cruise missiles and UAS. The Houthis' maritime campaign has threatened Gulf tanker routes. The design basis is drawn from events that have already occurred.

The DBA-OIL framework is a companion to the DBA-GAS framework, which addresses the natural gas sector under the same analytical architecture. Together, they provide the design basis for the petroleum and gas sector that federal regulators, industry operators, and allied defense planners need to assess, prepare for, and mitigate the consequences of military action against energy infrastructure.

1. Introduction

1.1 Purpose

This document establishes the Design Basis and Design Basis Accident — Military Action (DBA-MA) framework for the petroleum sector. It serves two related purposes.

First, it introduces the Design Basis concept to the petroleum sector. It establishes the methodology for identifying and characterizing bounding events — the events that petroleum sector facilities must be designed to withstand. The non-adversarial Design Basis defined in Section 4 establishes the baseline resilience posture against which all subsequent analysis is measured.

Second, it extends that Design Basis to the military-action threat tier. Section 5 introduces the four characteristics of military action that distinguish it from non-adversarial bounding events and explains why these characteristics require an extended analytical framework.

1.2 Scope and Applicability

This document governs the DBA-OIL series. It applies to petroleum sector infrastructure, including upstream crude oil production (onshore and offshore), petroleum refining and processing, crude oil and product pipelines, marine terminals and bulk liquid storage, and associated control systems. The DBA-OIL series primarily addresses the U.S. petroleum sector, with reference to allied and international contexts in which the consequences of military action extend beyond domestic boundaries.

Natural gas production, processing, and transmission are addressed in the DBA-GAS series. Where petroleum and natural gas infrastructure intersect, this document addresses the consequence chain as it originates in the petroleum sector, without displacing the DBA-GAS analysis of the gas-sector consequences of the same event.

1.3 Relationship to Governing Documents

This document is issued under the DBA-MA Parent Framework (IAEA L1 v1.4). DBA-EN-SF also governs this document: Energy Sector Design Basis Framework (v0.7), which sanctioned SDD as a series-standard metric at §4.4. The parent framework establishes the governing threat taxonomy, consequence analysis methodology, and framework architecture. Facility-class documents in the DBA-OIL series are issued under this sector framework and inherit the threat taxonomy, bounding methodology, and analytical structure established here.

1.4 How to Use This Document

A Design Basis answers a specific engineering question: what is the worst credible event this facility or system must be built to survive? Once the envelope is defined, the engineering question becomes tractable: which facility design, operating procedures, and recovery

architecture constitute adequate protection against the bounding event? The DBA-MA extension asks the harder version: what is the worst credible event when the initiating condition is deliberate military action?

Readers primarily interested in the military-action analysis may proceed directly to Section 5 after reading Section 3. Readers responsible for facility-class document development should read the full document, as the sector framework establishes the analytical inheritance that all facility-class documents carry.

2. Sector Profile

2.1 Sector Structure

2.1.1 Upstream: Crude Oil Production

Upstream infrastructure includes producing wells, wellheads, gathering pipelines, and field separation equipment for both onshore and offshore production. The United States is the world's largest crude oil producer, with major producing basins concentrated in the Permian Basin (Texas and New Mexico), the Gulf of Mexico outer continental shelf, the Bakken formation (North Dakota), and the Eagle Ford and Anadarko basins. Offshore Gulf of Mexico production accounts for approximately fifteen percent of total U.S. crude production and is concentrated on a relatively small number of major platforms and deepwater installations.

2.1.2 Midstream: Refining and Processing

Petroleum refining is the operationally critical and most consequential layer of the petroleum sector. The United States has approximately 130 operating refineries with a combined crude distillation capacity of approximately 18 million barrels per day. Refinery capacity is geographically concentrated: the Gulf Coast (PADD 3) accounts for approximately 45 percent of total U.S. refining capacity, creating a regional concentration vulnerability analogous to the Gulf Coast's role in natural gas production and processing.

2.1.3 Midstream: Pipelines and Transport

Crude oil trunk pipelines move crude from producing regions to refineries. Product pipelines distribute refined products from refineries to distribution terminals and end markets. The Colonial Pipeline system — the largest refined product pipeline in the United States — carries approximately 100 million gallons per day of gasoline, diesel, and jet fuel from Gulf Coast refineries to markets along the Eastern Seaboard. The 2021 ransomware attack on Colonial demonstrated that a non-state cyber actor could produce regional fuel supply disruptions with national visibility; the DBA-MA design basis addresses the state-actor version of the same attack with kinetic amplification.

2.1.4 Downstream: Terminals, Storage, and Distribution

Petroleum product terminals receive product from pipelines and marine vessels, store it in bulk liquid tanks, and distribute it by truck, rail, or local pipeline to retail and commercial customers. The Strategic Petroleum Reserve (SPR) — approximately 350 million barrels of crude oil stored in salt caverns on the Gulf Coast — is the primary federal emergency supply mechanism for crude oil shortages. Its geographic concentration on the Gulf Coast means it shares some of the same vulnerability exposure as the refinery capacity it is intended to support.

2.2 Regulatory Authority

The petroleum sector operates under a fragmented regulatory structure with jurisdiction divided among safety, environmental, economic, and security authorities. PHMSA holds pipeline safety jurisdiction under 49 CFR Parts 192 and 195. EPA holds environmental response jurisdiction. BSEE holds offshore production safety jurisdiction. FERC holds limited economic jurisdiction over oil pipelines. TSA holds surface transportation security authority and has issued mandatory cybersecurity directives following the Colonial Pipeline incident; physical security requirements for pipelines remain voluntary. CISA provides sector risk management support with advisory authority only.

2.3 Strategic Significance

Military logistics dependency. Petroleum products — jet fuel for aircraft, diesel for ground vehicles and naval vessels, and marine fuel for the logistics fleet — are the operational fuel of military power projection. A sustained disruption of domestic refining capacity that degrades jet fuel and diesel production directly constrains the military operational tempo at precisely the moment when military capacity is most needed.

Transportation and commerce dependency. The U.S. economy moves on petroleum products. Trucking, aviation, and rail freight depend on diesel and jet fuel from domestic refineries. A sustained refinery outage on the Gulf Coast produces supply shortfalls that cascade through freight costs, agricultural supply chains, and manufacturing logistics within days.

Allied energy security dependency. U.S. petroleum exports — crude oil, refined products, and natural gas liquids — are a component of the allied energy security architecture. A sustained disruption of Gulf Coast refining and export terminal capacity affects allied nations that depend on U.S. product exports as a marginal source of supply.

3. Design Basis: Petroleum Sector

3.1 The Concept and Its Origins

The Design Basis concept was formalized in nuclear safety engineering in the 1960s and 1970s as the foundation of reactor safety analysis. The concept's extension to the petroleum sector is a direct application of the same analytical logic. The petroleum sector presents the same foundational challenge: a complex system with many possible failure modes, finite design and investment resources, and a need to identify the bounding cases that drive hardening and resilience requirements.

3.2 Application to the Petroleum Sector

The Design Basis for the petroleum sector is defined at two levels. At the facility level, the Design Basis addresses the bounding events for individual asset classes. At the system level, it addresses the bounding events for the petroleum supply network as an integrated system.

A key distinction from the natural gas sector: the petroleum sector's cascade consequences are primarily economic and logistical rather than immediate life-safety. A refinery outage does not immediately threaten lives the way a heating gas supply disruption during a winter storm does. The consequence timeline is longer, the immediate life-safety consequence is lower, and the recovery mechanisms — including the SPR and international crude markets — are more developed.

The Design Basis serves a second function beyond establishing engineering requirements: it is a gap-finding instrument for policy, authority, and institutional coordination. Defining precisely what the sector must withstand makes visible the delta between that requirement and what existing regulatory frameworks actually compel.

4. Design Basis Accidents: Non-Adversarial Envelope

This section constitutes the DBA layer — the second stage of the DB → DBA → DBA-MA sequence. Non-adversarial bounding events fall into four categories: process safety events, natural hazard events, operational events, and cascading failure events.

4.1 Bounding Event Methodology

A bounding event is identified through a four-step process of hazard identification, consequence analysis, credibility assessment, and bounding selection. This methodology is established in the DBA-GAS Sector Framework and inherited here without repetition. The petroleum sector's application of the methodology yields different bounding events than the gas sector's because the hazard profile, consequence chains, and recovery mechanisms differ.

4.2 Design Basis Adequacy Standards

Supply continuity for priority products. The system must maintain the supply of priority petroleum products — military fuel grades, diesel for critical freight and emergency services, and heating oil for residential customers in regions dependent on petroleum heat — within defined supply-stress limits following the bounding event.

Recovery capability. The system must possess the redundancy, spare equipment inventory, and logistics capability needed to restore normal operations within a defined recovery window.

Environmental consequence limitation. Adequacy standards must include limits on environmental consequences, particularly for offshore production and marine terminal scenarios, where damage to the aquatic environment compounds the supply and economic consequences of the initiating event.

4.3 Process Safety Events

4.3.1 Major Refinery Fire and Explosion

Initiating event: Uncontrolled hydrocarbon release at a major process unit resulting in fire or explosion that damages the process unit and potentially adjacent units. The 2005 Texas City refinery explosion is the reference event. Consequence envelope: Physical destruction of one or more process units; loss of production across all product streams; recovery timeline of six to eighteen months for major unit reconstruction. Adequacy standard: Process safety management systems under OSHA 29 CFR 1910.119; catastrophic unit loss must not simultaneously disable fire suppression capability.

4.3.2 Large Crude Oil Spill — Pipeline

Initiating event: Rupture of a major crude oil trunk pipeline resulting in a large-volume crude oil release to the surrounding environment. The 2010 Enbridge Line 6B rupture in Marshall,

Michigan, is a reference event. Consequence envelope: Crude oil release to soil and water; environmental remediation costs and timelines that significantly exceed the costs of physical pipeline repair; supply curtailment to refineries served by the affected pipeline during the isolation and repair period. Adequacy standard: Block valve spacing and actuation capability must limit release volume; alternative crude supply routing or refinery crude flexibility must sustain refinery operations during the repair window.

4.3.3 Offshore Platform Blowout

Initiating event: Loss of well control at an offshore production platform resulting in an uncontrolled release of crude oil and associated gas. The Deepwater Horizon blowout (2010) is the reference event. Consequence envelope: Total production loss from the affected platform; environmental damage across the affected marine area; potential regulatory moratorium affecting other installations. Adequacy standard: BSEE well control regulations; offshore production facilities must have well control capability sufficient to achieve well shut-in within a defined time limit.

4.4 Natural Hazard Events

4.4.1 Hurricane Disruption of Gulf Coast Refining

Initiating event: A major hurricane (Category 3 or above) making landfall on or traversing the Gulf Coast, causing simultaneous physical damage, flooding, and extended power outages across the concentrated Gulf Coast refinery cluster. Hurricanes Katrina and Rita (2005) are the reference events. Consequence envelope: Physical damage to refinery process units; extended power outages affecting restart; damage to road and rail infrastructure. The bounding case involves a direct strike on the Houston Ship Channel/Beaumont refinery cluster.

4.4.2 Seismic Event — West Coast Refineries

Initiating event: A significant seismic event (magnitude 6.5 or greater) affecting the West Coast refinery concentration in the Los Angeles Basin or San Francisco Bay Area. Consequence envelope: Physical damage to refinery process units; loss of utility power; limited alternative supply given minimal pipeline interconnection to other refining regions. Adequacy standard: West Coast refinery infrastructure must meet seismic design standards appropriate to the facility's location on the applicable fault hazard map.

4.5 Operational Events

4.5.1 Control System Failure — Refinery

Initiating event: Failure or loss of availability of the distributed control system (DCS) governing a major refinery's process operations, requiring emergency shutdown of process units and transition to manual operations. Adequacy standard: Refinery control systems must have

redundant DCS architecture and manual control capability for emergency shutdown of each process unit.

4.5.2 Major Terminal Fire

Initiating event: Ignition of a petroleum product storage tank at a major product terminal, resulting in a tank fire that threatens adjacent tanks and terminal infrastructure. A boilover event represents the bounding case. Adequacy standard: Terminal fire suppression systems must be sized for the tank diameter and product type in accordance with NFPA 11 and API standards. Impoundment capacity must contain the contents of the largest tank plus firefighting water without offsite release.

4.6 Cascading Failure Events

4.6.1 Refinery-to-Transport Cascade

Initiating event: An event that simultaneously disables a major refinery cluster and the product pipeline serving its primary market, eliminating both production and transport redundancy for an extended period. Consequence envelope: The refinery-to-transport cascade is the highest-consequence non-adversarial event in the petroleum sector's Design Basis. Adequacy standard: Regional product supply resilience must be assessed at the system level, not only the facility level.

4.7 Governing Consequence Metric — Supply Delivery Duration (SDD)

The petroleum sector's pipeline transport facility class (DBA-OIL-FC4) is governed by the Supply Delivery Duration (SDD) metric: the continuous interval of pipeline delivery curtailment from the initiating event to restoration of full throughput capacity at the governing delivery point. SDD is established in DBA-OIL-FC4 v0.2 §§1.3 and 4.2 and is sanctioned as a series-standard metric at the DBA-EN-SF level (§4.4, v0.7).

SDD is the petroleum sector analog to GSD (Generation Supply Duration) in the electric sector and PPSD (Pipeline Pressure Supply Duration) in the natural gas sector. All three metrics follow the same analytical structure: a Zone 1/2/3 acceptance criterion framework with T₁/T₂ calibration thresholds, where site-specific calibration is required at implementation. SDD is not additive across pipeline segments — where multiple segments are affected, the governing SDD is the longest single-segment critical path, not the cumulative sum.

The SDD metric does not apply to the refining, offshore production, or terminal and storage facility classes, which are governed by RT_HVLLC (for physical destruction of long-lead components) and facility-specific process safety metrics established in their respective instruments. The relationship between facility-class metrics is established at the DBA-EN-SF level and inherited by this framework.

5. The Military Action Extension

5.1 Why the Non-Adversarial Design Basis Is Insufficient

Each bounding event in Section 4 shares a characteristic that distinguishes it from the military-action threat tier: it is not chosen. Process upsets occur according to the probabilistic distribution of equipment failure modes. Hurricanes follow meteorological paths. The adversary does not operate within this framework. Military action against petroleum infrastructure introduces the four characteristics that require an extended analytical framework.

5.2 The Four Military Action Characteristics

5.2.1 *Intentionality*

The adversary chooses its targets. The Gulf Coast refinery concentration is not a vulnerability that an adversary discovers by accident — it is publicly documented, economically significant, and geographically concentrated. An adversary who has studied the sector knows that the Houston Ship Channel refinery cluster accounts for a material fraction of U.S. refined product capacity. Intentionality means the adversary will target the fluid catalytic cracker whose replacement lead time is longest, not the least consequential unit whose spares are pre-positioned.

5.2.2 *Persistence*

Military action does not affect the sector's recovery timeline. A refinery fire is extinguished; repairs begin; the facility restarts in months. Military action can be sustained across the recovery period. The contractor crews mobilizing to repair blast-damaged process units are operating in a threat environment. The supply chain delivering replacement compressors and heat exchangers is a target.

5.2.3 *Precision*

Modern state actors can execute precision attacks against specific facility components. In a petroleum refinery, precision targeting allows the adversary to select the highest-consequence unit: the fluid catalytic cracker, the alkylation unit, or the hydrogen plant. The Ukrainian experience is directly applicable: Russian strikes on Ukrainian petroleum refining infrastructure from 2022 through 2024 demonstrated that an adversary with precision-strike capability will target the specific process equipment whose loss maximizes disruption duration.

5.2.4 *Denial*

Military action can deny site access. The firefighting crews that would contain a process fire, the inspection teams that would assess damage and authorize restart, and the contractor crews that would execute repairs cannot access a facility under active military threat. The local sovereignty

principle — that protective systems must operate passively, without external power, communications, or operator action — applies to the petroleum sector.

5.3 The Extended Design Basis: DBA-MA

The DBA-MA extension builds on the non-adversarial Design Basis rather than replacing it. The DBA-MA bounding event is in every case more severe than its non-adversarial analog. The adversary selects the highest-consequence configuration, times the attack for maximum effect, coordinates simultaneous events that defeat single-scenario resilience mechanisms, and denies the access on which recovery depends.

5.4 The Command-Broker Architecture and the Bright Line

The command-broker architecture is the Design Basis requirement for all ML and Gen AI components whose outputs can affect physical process parameters in petroleum sector control systems. The Bright Line is the Bright Line as defined in FD-BL (Foundational Definition — The Bright Line): a threshold on individual actions. Under FD-BL §4.2–§4.3, an action on a physical process parameter is above the Bright Line when its incorrect performance could produce a consequence at or above the facility's severity threshold, and below it when it could not — placement follows the consequence of the action, not whether the component is characterized as advisory or resides in a command layer. The advisory / physical-command-layer seam remains the natural place to build enforcement, but it is not the definition. Above-Bright-Line actions require a human command broker; the boundary is enforced by architecture, not by procedure or operator discipline. DBA-OIL facility-class instruments inherit this definition by reference to FD-BL and identify, for each facility class, the specific physical process parameters whose actions fall above the Bright Line and the command-broker architecture that enforces authorization for them.

The canonical cross-sector instrument governing this architecture is the Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector v0.1). For the petroleum sector, the highest-criticality tier governs all applications that control or influence process unit operations, fired heater management, safety instrumented system parameters, pressure relief logic, and emergency shutdown systems. The Command Broker is the human operator who decides based on process observables and Gen AI advisory output. A vendor characterization of embedded Gen AI as providing “autonomous process optimization” or “direct control integration” describes a capability that is inconsistent with Design Basis compliance in the highest-criticality petroleum sector applications.

6. DBA-MA Threat Framework for the Petroleum Sector

6.1 Threat Actor Taxonomy

Tier 1 — State Actor. Nation-states with advanced persistent threat capability, precision strike capability, and the demonstrated willingness to employ combined cyber, kinetic, and influence operations against petroleum infrastructure. Russia, Iran, and China present credible Tier 1 threats to U.S. petroleum sector infrastructure.

Tier 2 — Near-State Proxy. Paramilitary and hybrid warfare organizations operating under state direction. The Houthi maritime campaign demonstrated Tier 2 capability against energy-sector targets at sea. Applied to Gulf Coast marine terminals, offshore platforms, or refinery waterfront access points, the Houthis' operational model is directly relevant.

Tier 3 — Sophisticated Non-State Actor. Criminal organizations and ideologically motivated actors. The Colonial Pipeline ransomware incident is the Tier 3 reference event: significant consequences, but substantially less severe than a Tier 1 or Tier 2 attack against the same infrastructure.

6.2 Attack Modality Taxonomy

Kinetic Direct. Physical destruction of refinery process units, pipeline infrastructure, storage tanks, or marine terminal facilities using conventional explosives, precision-guided munitions, or UAS-delivered munitions.

Cyber-Kinetic Combined Arms. Integration of ICS/SCADA compromises with kinetic action. DCS compromise can manipulate process conditions to create process upsets that amplify the physical damage of a concurrent kinetic strike.

Supply Chain and Logistics Denial. Interdiction of the supply chains for critical replacement equipment to extend recovery timelines after a kinetic attack. A Tier 1 adversary that understands heat exchanger and compressor replacement lead times can design supply chain interdiction operations that sustain disruption long after physical damage assessment and repair have begun.

6.3 Bounding Logic

The Design Basis Threat for the petroleum sector is: Tier 1 cyber pre-positioning with Tier 2 kinetic execution; combined cyber-kinetic modality; target selection for maximum multi-product supply impact; timing at peak demand conditions; and simultaneity within a window that exceeds the sector's adaptive capacity.

7. Consequence Interaction and Cross-Sector Dependencies

7.1 Military Logistics Dependency

The petroleum sector's most consequential cross-sector dependency under military-action conditions is the military logistics fuel supply chain. U.S. military aircraft, ground vehicles, and naval vessels depend on jet fuel (JP-8/Jet-A), diesel, and marine distillate produced by domestic refineries. A sustained disruption of Gulf Coast refining capacity that reduces jet fuel and diesel production constrains military operational tempo at precisely the moment when military capacity is most needed.

7.2 Transportation and Commerce Cascade

Refined product supply disruptions cascade through transportation and commerce. Trucking fuel price spikes and diesel rationing affect freight costs and delivery schedules across the entire supply chain. Aviation fuel shortfalls affect both commercial aviation and military airlift. Agricultural diesel shortfalls during planting or harvest seasons have implications for food security.

7.3 Environmental Consequence as a Strategic Amplifier

Environmental consequences of petroleum sector attacks — crude oil spills, product releases, refinery fire-suppression runoff — amplify the strategic effect of military action in ways that do not apply to the nuclear or gas sectors. An adversary who designs an attack to maximize environmental consequence achieves a disruption multiplier through the regulatory and remediation burden rather than through physical damage alone.

7.4 Market Consequence and Price Signal Cascade

Petroleum markets respond to supply-disruption signals faster than physical supply chains do. A credible attack on Gulf Coast refining infrastructure produces immediate price spikes in crude oil, gasoline, and diesel futures markets. An adversary who understands this dynamic can design attacks to maximize the market signal to achieve economic disruption through price effects that outrun the actual physical supply impact.

8. Protective Action and Recovery Architecture

8.1 Threat Detection Framework

Elevated threat indicators include: Intelligence Community reporting of adversary interest in petroleum refinery or terminal infrastructure; anomalous ICS network activity at refinery DCS systems; physical surveillance of refinery fence lines or marine terminal access points; unusual acquisition patterns for commercially available explosives or process engineering documentation.

Imminent threat indicators include: confirmed physical intrusion at refinery or terminal perimeter; simultaneous DCS anomalies at multiple refineries within a single regional cluster; anomalous pressure readings at multiple pipeline nodes within a single system within a short time window.

8.2 Decision Authority Framework

Facility operators retain authority to initiate emergency shutdown, process unit isolation, and site evacuation without external authorization. Sector coordination through the ONE-ISAC and the Oil and Natural Gas Subsector Coordinating Council activates mutual aid. Federal ESF-12 activation under DOE leadership coordinates federal support for supply restoration. SPR drawdown for a crude supply emergency requires the Secretary of Energy or Presidential authority. Emergency response under OPA 90 requires FOSC activation by the USCG or EPA.

8.3 Federal Coordination Triggers

The following events constitute Federal Coordination Triggers — thresholds that require mandatory federal coordination and, where authority exists, federal response action: confirmed coordinated attack on two or more petroleum refinery or terminal facilities within a single regional cluster within 48 hours; refined product supply disruption affecting military logistics fuel supply contracts above a defined threshold; ICS compromise with confirmed or strongly suspected nation-state attribution at two or more petroleum sector facilities; confirmed crude oil or product release to a navigable waterway resulting from circumstances consistent with adversary action; and confirmed fatalities at petroleum sector facilities in circumstances consistent with adversary action.

8.4 Recovery Sequencing Principles

Phase 1 — Immediate stabilization. Emergency shutdown of affected process units. Fire suppression activation. Pipeline isolation at nearest block valves. SPR drawdown authorization for crude supply emergency. Military fuel reserve activation for priority military consumers.

Phase 2 — Short-term bridging. Undamaged refinery capacity maximization. Product import surge from Atlantic Basin refineries. SPR crude drawdown to supplement refinery feedstock. Demand reduction measures for civilian fuel supply.

Phase 3 — Medium-term restoration. Refinery damage assessment and repair scope definition. Specialty contractor mobilization. Replacement equipment procurement. Sequential process unit restart following safety case re-validation and regulatory inspection. Supply chain monitoring for adversary interdiction of replacement equipment.

The local sovereignty principle applies to Phase 1 protective actions. Emergency shutdown systems, pressure relief devices, and fire and gas detection systems must operate passively, without external power, communications, or operator action.

9. Facility Class Register and Inheritance Model

9.1 The Inheritance Model

Facility-class documents in the DBA-OIL series inherit the sector framework’s threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards. They do not repeat this material. Each facility-class instrument must include a section explicitly identifying which elements are inherited from this document unchanged, which elements are inherited with facility-class-specific adaptation, and which elements are entirely locally defined.

9.2 Facility Class Placement Rationale

The petroleum sector is organized around the physical pathway from wellhead through refining to distribution. Facility classes in the DBA-OIL series are defined by their position in that pathway and by the consequence chain their loss initiates. Petroleum refining (FC1) is placed first because refinery outage is the highest-consequence single-facility event in the domestic petroleum sector. Offshore crude production (FC2) is placed second because wellhead outage reduces the feedstock available to the refinery system. Terminals and storage (FC3) address the intermediate logistics layer, excluding the strategic marine terminals carved out as FC5. Crude and product transport (FC4) addresses the pipeline network; SDD is the governing metric for this facility class. Marine crude and product terminals (FC5) are placed last as a promoted facility class, carved from the marine scope of FC3. The boundary between FC3 and FC5 is drawn on strategic significance rather than facility capability: a marine terminal is registered as FC5 when it carries a defense-fuel logistics commitment, an allied or strategic export commitment, or is a designated strategic crude-import chokepoint; all other marine and onshore terminals and bulk storage remain in FC3. The promotion reflects the determination that strategic marine terminals carry a consequence dimension — import dependency and allied-export commitment — that the general FC3 logistics treatment does not capture.

9.3 Facility Class Register

Designation	Facility Class	NIPP Sector	Status	Primary Regulatory Anchor
DBA-OIL-FC1	Petroleum refining	Energy	Active — v1.3	EPA RMP (40 CFR Part 68); OSHA PSM (29 CFR 1910.119); NERC CIP-014 (where applicable)
DBA-OIL-FC2	Offshore crude production — platforms and associated pipeline	Energy	Active — v0.1	BSEE 30 CFR Parts 250/254; USCG 33 CFR; PHMSA 49 CFR Part 195
DBA-OIL-	Crude and product	Energy	Active —	EPA SPCC (40 CFR Part

FC3	terminals and storage — general onshore and non-strategic marine		v0.1	112); USCG 33 CFR; PHMSA 49 CFR Part 195; TSA Pipeline Security Directives
DBA-OIL-FC4	Crude and product transport — trunk and product pipelines (SDD metric)	Energy	Active — v0.2	PHMSA 49 CFR Part 195; TSA Pipeline Security Directives; EPA SPCC
DBA-OIL-FC5	Marine crude and product terminals — strategic import/export (carved from FC3)	Energy	Active — promoted; companion DBA-OIL-MA-SF v0.1	USCG/MTSA 33 CFR Parts 104 and 105; PHMSA 49 CFR Part 195; EPA SPCC (40 CFR Part 112)

9.4 Development Sequence

FC2 (Offshore Crude Production) is the next facility-class instrument to be developed, driven by the maritime threat profile and the remote-access constraints that distinguish offshore operations from onshore refining. FC3 (Terminals and Storage) follows, addressing the tank farm fire, spill, and supply chain disruption consequence chains. FC4 (Crude and Product Transport) follows FC3; Colonial Pipeline is the primary reference event. All four original facility-class instruments are at first issue. The promoted FC5 (Marine Crude and Product Terminals) is registered and pending dedicated instrument development; it is currently covered by DBA-OIL-MA-SF v0.1 and the marine content carved from FC3.

10. Gaps and Open Items

10.1 Regulatory Gap: Absence of Mandatory Physical Security Standards

The petroleum sector lacks mandatory physical security standards equivalent to the electricity sector's NERC CIP-014. TSA's pipeline security guidelines are voluntary for physical security; TSA's mandatory cybersecurity directives following the Colonial Pipeline incident address only the cyber dimension. No regulatory mechanism compels the implementation of the Design Basis established in this document.

10.2 Gulf Coast Concentration and Resilience Planning

The concentration of Gulf Coast refineries is the defining structural vulnerability of the U.S. petroleum sector. No federal program currently requires refiners, pipeline operators, or federal regulators to assess the system-level consequences of a simultaneous attack on the Gulf Coast cluster.

10.3 SPR Positioning and Military-Action Scenarios

The Strategic Petroleum Reserve was designed for scenarios of international supply disruption, not domestic infrastructure attacks. Its geographic concentration on the Gulf Coast means that the most consequential domestic attack scenario may also degrade SPR drawdown and distribution capabilities.

10.4 Environmental Consequences and Military-Action Response

OPA 90 facility response plans address worst-case discharge scenarios under accidental conditions. They do not address the military-action scenario, in which response resources may be unable to access the site and the simultaneous occurrence of fires, explosions, and security threats creates a multi-hazard response environment that exceeds the planning basis of current environmental response frameworks.

Revision History

Version	Date	Author	Description
1.0	April 2026	Tim Roxey	Initial version. Establishes the Design Basis and DBA-MA framework for the petroleum sector. Introduces the non-adversarial Design Basis methodology and bounding event categories. Defines the four characteristics of military action and their implications. Establishes sector-level threat taxonomy, consequence interaction framework, and facility-class instrument architecture. DBA-OIL-FC1 (Petroleum Refining) issued concurrently.
1.2	May 2026	Tim Roxey	Section 9 — “Facility Class Instruments” replaced with full “Facility Class Register and Inheritance Model” architecture (Sections 9.1–9.4). Existing inheritance model prose retained and extended with a mandatory inheritance-map requirement. New Sections 9.2 (Placement Rationale), 9.3 (five-column register table), and 9.4 (Development Sequence) added.
1.3	June 2026	T. Roxey	Section 5.4 (The Command-Broker Architecture and the Bright Line) added. Establishes the three-tier criticality framework inherited from CB-Framework_Cross_Sector v0.1; defines Command Broker requirement for highest-criticality petroleum sector applications; establishes vendor-embedded ML and Gen AI assessment requirements. Section 8.3 renamed from “Bright Line Criteria” to “Federal Coordination Triggers”; body text updated accordingly. “Bright Line” is reserved exclusively for the ML/AI governance boundary throughout this document.
1.4	June 2026	T. Roxey	Housekeeping revision. “Brightline” corrected to “Bright Line” (two words) throughout §5.4 heading and body, consistent with series-wide convention. FC1 register entry updated from Active — v1.0 to Active — v1.2. v1.3 revision history entry added (previously undocumented). No substantive content changes.
1.5	June 2026	T. Roxey	Housekeeping revision. FC2 register entry updated from Anticipated to Active — v0.1 (DBA-OIL-FC2: Offshore Crude Production). FC3 register entry updated from Anticipated to Active — v0.1 (DBA-OIL-FC3: Crude and Product Terminals and Storage). No substantive content changes.
1.6	June 2026	T. Roxey	Housekeeping revision. FC4 register entry updated from Anticipated to Active — v0.1 (DBA-OIL-FC4: Crude and Product Transport — Trunk and Product Pipelines). All four DBA-OIL facility class instruments are now at first issue. No

			substantive content changes.
1.7	June 2026	T. Roxey	Section 4.7 (Governing Consequence Metric — Supply Delivery Duration) added. Registers SDD as the governing consequence metric for the DBA-OIL-FC4 pipeline transport facility class, sanctioned at the DBA-EN-SF level (§4.4, v0.7). States SDD non-additivity rule and Zone 1/2/3 acceptance criterion structure. Notes inapplicability to refining, offshore production, and terminal/storage facility classes. Follows the same sanctioning pathway as PPSD (established at FC level, registered upward to sector framework and DBA-EN-SF). FC4 register entry updated from Active — v0.1 to Active — v0.2.
1.8	June 2026	T. Roxey	Marine/strategic terminal classification. The marine scope of FC3 is split: strategic marine crude and product terminals are carved out and registered as DBA-OIL-FC5 in §9.3, with the FC3/FC5 boundary drawn on strategic significance (§9.2). FC3 scope narrowed to general onshore and non-strategic marine terminals and storage. Promotion asserted in DBA-OIL-MA-SF v0.1 and now registered here. Parallels the gas LNG promotion in DBA-GAS-SF v1.7; register the two together at the parent. No other substantive change.